

decrease in temperature, output power could easily go past the absolute maximum value.

The circuit topology in the case of constant-current mode of operation is usually configured around a current-sensing element that continuously senses the drive current and produces a proportional voltage. This is then compared with a reference voltage representing the desired value of drive current to generate an error signal. The error signal, after suitable conditioning, is fed back to restore the drive current to the desired value. Fig. 1 shows the basic schematic arrangement of constant-current drive circuit.

Practical circuits with in-built protection features are far more complex. Typical constant-current drive circuits and related design issues are discussed in the following paragraphs.

There are applications where constant output power is more relevant. In such a situation, direct dependence of output power on drive current can be exploited to maintain a constant output power. Even when a laser diode is driven by a constant current, the heat dissipated at the laser-diode junction leads to rise in temperature and hence fall in output power. Increase in temperature could be because of absence of any active temperature control mechanism or even inadequacy of heat sink in the presence of active temperature control. This reduction in output power can be compensated by increasing the drive current. The current source can be designed in such a way that it adjusts the drive current in a feedback mode to maintain a constant output power instead of maintaining a constant drive current. Fig. 2 shows the basic circuit topology. The basic circuit topology of constant-output-power drive circuit is similar to that of constant-current drive circuit, except for the nature of sense signal, which in the present case is a photo current proportional to the output power.

Constant-output-power drive circuit needs to be operated with an absolute current limit to prevent any



Fig. 3: Laser diode with integral photodiode

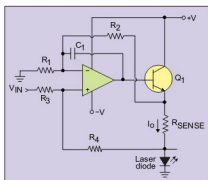


Fig. 4: Constant-current laser diode drive circuit for grounded load

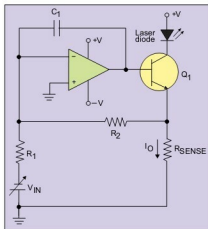


Fig. 5: Constant-current laser diode drive circuit for floating load

thermal runaway problem caused by uncontrolled increase in drive current. Laser diode modules with an integral photodiode (Fig. 3) facilitate constant-output-power operation, though noise intrinsic to the integral photodiode manifests itself in the form of noisy and unstable output in the case of constant-output-power mode of operation. The device usu-

ally has three terminals, including either anode or cathode of laser diode, either anode or cathode of photodiode and a common terminal obtained by connecting the remaining terminals of laser diode and photodiode. In some cases, all four terminals, two of laser diode and two of photodiode, are brought out.

### Laser-diode drive circuit: constant-current mode

Fig. 4 shows the basic constant-current source circuit for grounded-laser-diode configuration. Laser-diode current is sensed differentially by measuring voltage across  $R_{SENSE}$  wired in series with laser diode. Laser diode current in this case can be computed from  $I_O = V_{IN} \times R_1 / (R_3 \times R_{SENSE})$ , where  $R_1 = R_2$  and  $R_3 = R_4$ .  $C_1$  is the compensation capacitor connected for stable operation.  $V_{IN}$  may be derived from a band-gap reference. In case the control voltage  $V_{IN}$  is of negative polarity, it is connected to  $R_1$  and  $R_3$ , and is grounded instead. A digitally-controlled current source may use a voltage output digital-to-analogue converter to generate  $V_{IN}$ .

Similar circuit for a floating load is shown in Fig. 5. Laser-diode current in this case can be computed from  $I_O = V_{IN} \times R_2 / (R_1 \times R_{SENSE})$ .

The voltage-controlled constant-current sources discussed so far provide constant-current drive to the laser diode, provided the DC supply voltage, the voltage at the emitter terminal and the resistance in the emitter lead were all constant. It would be a reasonably good assumption if the source voltage were derived from a precision band-gap reference and the resistors used had stability specifications equal to or better than the desired level of current stability. But when it comes to achieving a higher level of current stability, say  $\pm 10\text{ppm}$  or better, a feedback loop that in situ samples the diode current and applies a correction in the case of drift in drive current becomes essential. A negative feedback loop of this kind would reduce the drift or error in the drive current by a factor that equals the loop gain.